

Exercise 6.1. Let $e_0, \dots, e_{m-1}$ a list of vectors in V , such that

$$\|a_0 e_0 + \dots + a_{m-1} e_{m-1}\|^2 = |a_0|^2 + \dots + |a_{m-1}|^2$$

for any $a_0, \dots, a_{m-1} \in V$. Show that $e_0, \dots, e_{m-1}$ is an orthonormal basis.

Proof. For any $i = 0, \dots, m-1$, we have $\|e_i\|^2 = |1|^2$ if we pick $a_i = 1$ and others = 0.

For any $i, j = 0, \dots, m-1$ and $i \neq j$, we have $\|a_i e_i + a_j e_j\|^2 = |a_i|^2 + |a_j|^2$. Then

$$\begin{aligned} & \|a_i e_i + a_j e_j\|^2 \\ &= \langle a_i e_i + a_j e_j, a_i e_i + a_j e_j \rangle \\ &= \langle a_i e_i, a_i e_i \rangle + \langle a_j e_j, a_j e_j \rangle + \langle a_i e_i, a_j e_j \rangle + \langle a_j e_j, a_i e_i \rangle \\ &= a_i \bar{a}_i \langle e_i, e_i \rangle + a_j \bar{a}_j \langle e_j, e_j \rangle + a_i \bar{a}_j \langle e_i, e_j \rangle + a_j \bar{a}_i \langle e_j, e_i \rangle \\ &= a_i \bar{a}_i + a_j \bar{a}_j + a_i \bar{a}_j \langle e_i, e_j \rangle + a_j \bar{a}_i \langle e_j, e_i \rangle \\ &= |a_i|^2 + |a_j|^2 + a_i \bar{a}_j \langle e_i, e_j \rangle + a_j \bar{a}_i \overline{\langle e_i, e_j \rangle} \\ &= |a_i|^2 + |a_j|^2 \end{aligned}$$

that means $a_i \bar{a}_j \langle e_i, e_j \rangle + a_j \bar{a}_i \overline{\langle e_i, e_j \rangle} = 0$. Pick $a_i = a_j = 1$ we have $2 \operatorname{Re} \langle e_i, e_j \rangle = 0$ and pick $a_i = i, a_j = 1$ we have $-2 \operatorname{Im} \langle e_i, e_j \rangle = 0$. $\square$

Exercise 6.2.

- Let $\theta \in R$, show that:

$$(\cos \theta, \sin \theta), (-\sin \theta, \cos \theta) \quad \text{and} \quad (\cos \theta, \sin \theta), (\sin \theta, -\cos \theta)$$

are orthonormal basis of R^2 .

- Show that any orthogonal basis of R^2 is in one of such form.

Proof.

- They are normal cause $\cos^2 \theta + \sin^2 \theta = 1$, and orthogonal cause $-\cos \theta \sin \theta + \sin \theta \cos \theta = \cos \theta \sin \theta - \sin \theta \cos \theta = 0$.
- Any vector with length 1 in R^2 must in form of $(\cos \theta, \sin \theta)$. For any orthonormal basis, suppose it is $(\cos \theta, \sin \theta), (\cos \alpha, \sin \alpha)$, we may

write α as $\beta - \frac{\pi}{2}$, then the inner product of them is $\cos \theta \cos(\beta - \frac{\pi}{2}) + \sin \theta \sin(\beta - \frac{\pi}{2})$. Recall that $\sin \theta = \cos(\theta - \frac{\pi}{2})$ for any θ , then

$$\begin{aligned}
& \cos \theta \cos(\beta - \frac{\pi}{2}) + \sin \theta \sin(\beta - \frac{\pi}{2}) \\
&= \cos \theta \sin \beta + \sin \theta \cos(\beta - \pi) \\
&= \cos \theta \sin \beta + \sin \theta (-\cos \beta) \\
&= \cos \theta \sin \beta - \sin \theta \cos \beta \\
&= \sin(\theta - \beta) \\
&= 0
\end{aligned}$$

That means $\theta + \beta = k\pi$, thus $\theta + \alpha = k\pi - \frac{\pi}{2}$. If k is odd, then $\cos \alpha = \cos(k\pi - \frac{\pi}{2} - \theta) = \cos(\pi - \frac{\pi}{2} - \theta) = \sin(\pi - \theta) = -\sin(-\theta) = \sin \theta$ and $\sin \alpha = \sin(k\pi - \frac{\pi}{2} - \theta) = \sin(\pi - \frac{\pi}{2} - \theta) = \sin(\frac{\pi}{2} - \theta) = \sin(-(\theta + \frac{\pi}{2})) = -\sin(\theta + \frac{\pi}{2}) = -\cos \theta$. Similar to the case that k is even.

□

Exercise 6.9. Suppose $e_0, \dots, e_{m-1}$ is the result of applying Gram-Schmidt Procedure to a linear independent list $v_0, \dots, v_{m-1}$. Show that $\langle v_k, e_k \rangle > 0$ for all k .

Proof. By definition, we have $e_k = \frac{f_k}{\|f_k\|}$ where $f_k = v_k - \frac{\langle v_k, f_0 \rangle}{\|f_0\|^2} f_0 - \dots$, thus $v_k = f_k + \frac{\langle v_k, f_0 \rangle}{\|f_0\|^2} f_0 + \dots = e_k \|f_k\| + \dots$, thus $\langle v_k, e_k \rangle = \|f_k\| > 0$. □

Exercise 6.10. Let $v_0, \dots, v_{m-1}$ is linear independent list in V . Explain why the result of applying Gram-Schmidt Procedure on $v_0, \dots, v_{m-1}$ is the only orthonormal list such that for all $k = 0, \dots, m-1$, $\langle v_k, e_k \rangle > 0$ and $\text{span}(v_0, \dots, v_k) = \text{span}(e_0, \dots, e_k)$.

Proof. Let $e_0, \dots, e_{m-1}$ is the result of Gram-Schmidt procedure and $f_0, \dots, f_{m-1}$ is any orthonormal list holds the properties.

We induction on m :

- Base($m = 0$): We have $\text{span}(v_0) = \text{span}(e_0) = \text{span}(f_0)$, which means $f_0 = ke_0$, then we have $\langle v_0, f_0 \rangle = \langle v_0, ke_0 \rangle = \bar{k} \langle v_0, e_0 \rangle > 0$. By $\langle v_0, e_0 \rangle > 0$ we know $\langle v_0, e_0 \rangle$ is real, then $\bar{k}$ is real, thus k is real. Then by $\langle f_0, e_0 \rangle = \langle ke_0, e_0 \rangle = k$ and $\langle f_0, e_0 \rangle = \langle f_0, \frac{1}{k} f_0 \rangle = \frac{1}{k}$, we know $|k| = 1$. Then by $\bar{k} \langle v_0, e_0 \rangle > 0$ and $\langle v_0, e_0 \rangle > 0$, we know $k > 0$, thus $k = 1$, therefore $f_0 = e_0$.

- Ind($m = n + 1$): $e_0, \dots, e_{m-1} = f_0, \dots, f_{m-1}$ if they holds the property (positive inner product and span), and they hold by assumption. Then $\text{span}(e_0, \dots, e_{m-1}) = \text{span}(f_0, \dots, f_{m-1})$, and e_m, f_m are orthogonal to any vector in $\text{span}(e_0, \dots, e_{m-1})$. Since $\text{span}(f_0, \dots, f_m) = \text{span}(v_0, \dots, v_m) = \text{span}(e_0, \dots, e_m)$, we can write f_m in form of $c_0 e_0 + \dots + c_m e_m$, where $c_k = \langle f_m, e_k \rangle$, thus $f_m = c_m e_m$, as f_m is orthogonal to any vector in $\text{span}(e_0, \dots, e_{m-1})$, therefore $\text{span}(f_m) \in \text{span}(e_m)$. Then we can apply the step as Base case did to f_m and e_m , and then $f_m = e_m$. Thus $e_0, \dots, e_m = f_0, \dots, f_m$.

In fact, the proof doesn't involve any property of the result of Gram-Schmidt Procedure, so it says any two orthonormal list with those properties are equal, and the result of Gram-Schmidt Procedure holds those properties. $\square$

Exercise 6.13. *A $v_0, \dots, v_{m-1}$ list in V is linear dependent $\iff$ one vector of the result of Gram-Schmidt Procedure is 0.*

Proof.

- ($\Leftarrow$) Each f_k is a (non-trivial) linear combination of $f_0, \dots, f_{k-1}, v_k$, if $f_k = 0$ for some k , that means there is a (non-trivial) linear combination of $f_0, \dots, f_{k-1}, v_k$ produces 0, where $\text{span}(f_0, \dots, f_{k-1}) = \text{span}(v_0, \dots, v_{k-1})$, thus $f_k = 0$ implies $v_k \in \text{span}(v_0, \dots, v_{k-1})$, thus $v_0, \dots, v_k$ is linear dependent.
- ($\Rightarrow$) Let k minimum such that $v_0, \dots, v_k$ is linear dependent, thus $v_0, \dots, v_{k-1}$ is linear independent. We know f_k is a linear combination of $f_0, \dots, f_{k-1}, v_k$, where $\text{span}(f_0, \dots, f_{k-1}) = \text{span}(v_0, \dots, v_{k-1})$, thus $f_k \in \text{span}(f_0, \dots, f_{k-1}, v_k) = \text{span}(v_0, \dots, v_{k-1}, v_k) = \text{span}(v_0, \dots, v_{k-1}) = \text{span}(f_0, \dots, f_{k-1})$. Also, we know f_k is orthogonal to any previous f_{k-i} , thus if we write f_k in a linear combination of $f_0, \dots, f_{k-1}$, we will see all coefficients are 0, thus $f_k = 0$.

$\square$

Exercise 6.14. *Let V a real inner product vector space, and $v_0, \dots, v_{m-1}$ a linear independent list in V . Show that there are 2^m distinct orthonormal list $e_0, \dots, e_{m-1}$, such that:*

$$\text{span}(v_0, \dots, v_k) = \text{span}(e_0, \dots, e_k)$$

for any $k = 1, \dots, m$.

Proof. Binary.

We start from applying Gram-Schmidt Procedure on $v_0, \dots, v_{m-1}$, say $e_0, \dots, e_{m-1}$. Then for each e_k , we can replace it with $-e_k$, thus we have at least 2^m distinct orthonormal list holds the property.

For at most case, it should be done easily by 6.10. $\square$

Exercise 6.15. Suppose $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are inner product over V , such that $\langle u, v \rangle_1 = 0 \iff \langle u, v \rangle_2 = 0$. Show that there is $c > 0$ such that $\langle u, v \rangle_1 = c \langle u, v \rangle_2$ for any $u, v \in V$.

Proof. For any $v, w \in V$, we can write w in form of $av + u$ where $v \perp u$. Then $\langle v, w \rangle_1 = \langle v, av + u \rangle_1 = \bar{a} \langle v, v \rangle_1$ and $\langle v, w \rangle_2 = \langle v, av + u \rangle_2 = \bar{a} \langle v, v \rangle_2$, thus $\langle v, w \rangle_1 = \frac{\langle v, v \rangle_1}{\langle v, v \rangle_2} \langle v, w \rangle_2$. Similarly, we can write v in form of $aw + u$ where $w \perp u$, then $\langle v, w \rangle_1 = \langle aw + u, w \rangle_1 = a \langle w, w \rangle_1$ and $\langle v, w \rangle_2 = \langle aw + u, w \rangle_2 = a \langle w, w \rangle_2$, thus $\langle v, w \rangle_1 = \frac{\langle w, w \rangle_1}{\langle w, w \rangle_2} \langle v, w \rangle_2$. Thus $\frac{\langle v, v \rangle_1}{\langle v, v \rangle_2} = \frac{\langle w, w \rangle_1}{\langle w, w \rangle_2}$, that means the scalar between two different inner products of different vectors are the same. $\square$

Exercise 6.17. Let $F = \mathbb{C}$ and V is finite. Show that let $T \in \mathcal{L}(V)$, and the only eigenvalue is 1, and for any v , $\|Tv\| \leq \|v\|$, then $T = I$.

Proof. Since $F = \mathbb{C}$ and V is finite, we know there is a basis $v_0, \dots, v_{m-1}$ such that $\mathcal{M}(T)$ is upper-triangular. Then apply Gram-Schmidt Procedure on $v_0, \dots, v_{m-1}$, say $e_0, \dots, e_{m-1}$. It is easy to show that $\mathcal{M}(T, e_0, \dots, e_{m-1})$ is upper-triangular, as for any k , we have $\text{span}(e_0, \dots, e_k) = \text{span}(v_0, \dots, v_k)$ is invariant under T .

Then for any k , we know $\|Te_k\| \leq \|e_k\| = 1$, however, $\|Te_k\| \geq 1 \|e_k\| = 1$ (cause Te_k must be a linear combination of $e_0, \dots, e_k$, and we know the coefficients of e_k is 1, which is in the diagonal/the eigenvalue). Thus $\mathcal{M}(T, e_0, \dots, e_{m-1})$ is a diagonal matrix with all 1 in the diagonal, which is I . $\square$

Exercise 6.18. Let $u_0, \dots, u_{m-1} \in V$ a linear independent list. Show that there is $v \in V$ such that $\langle u_k, v \rangle = 1$ for all $k \in \{0, \dots, m-1\}$.

Proof. Induction on m .

- Base($m = 1$): Take $v = \frac{1}{\|u_k\|^2} u_k$.
- Ind($m = n+1$): By induction hypothesis, we have v such that $\langle u_k, v \rangle = 1$ for all $k \in \{0, \dots, n-1\}$. Then let $w \in V$ such that w is orthogonal

to $u_0, \dots, u_{n-1}$. this is possible by using Gram-Schmidt Procedure on $u_0, \dots, u_{m-1}$. And s such that $\bar{s} = \frac{1 - \langle u_{m-1}, v \rangle}{\langle u_{m-1}, w \rangle}$, then we have $\langle u_{m-1}, v + sw \rangle = \langle u_{m-1}, v \rangle + \langle u_{m-1}, sw \rangle = \langle u_{m-1}, v \rangle + \bar{s} \langle u_{m-1}, w \rangle = \langle u_{m-1}, v \rangle + (1 - \langle u_{m-1}, v \rangle) = 1$. Also, for any u_k where $k = \{0, \dots, n-1\}$, we have $\langle u_k, v + sw \rangle = \langle u_k, v \rangle + \langle u_k, sw \rangle = \langle u_k, v \rangle = 1$, since w is orthogonal to u_k . Thus $\langle u_k, v + sw \rangle = 1$ for all $k \in \{0, \dots, m-1\}$.

□